

Sonya Simkin

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA, USA

Aug 2024 – Dec 2025

M.S. Computer Science - Research Thesis

- Thesis: "Logical Relations for Noninterference with Cryptography"
- Advised by [Stephanie Balzer](#)

Carnegie Mellon University, Pittsburgh, PA, USA

Aug 2020 – May 2024

B.S. Computer Science

- Concentration: Principles of Programming Languages

RESEARCH EXPERIENCE

Principles of Programming Group [[Website](#)]

May 2024 – Dec 2025

Graduate Student

- Contributed to the development and mechanization of language-agnostic logical relations for prescribing session-typed protocols to heterogeneous systems in **Rocq**
- Defined logical relations for a language with encryption primitives using an information-flow control type system to prove possibilistic noninterference

Transportation, Bots, and Disability (TBD) Lab [[Website](#)]

Jun 2022 – Aug 2022

Research Assistant

- Implemented new features for a data labeling UI in **MATLAB App Designer**, including the deletion, addition, editing, and relabeling of data points projected onto a video
- Co-authored a publication that provides a system for the collection, semi-autonomous labeling, and correction of pedestrian data for machine-learning and robotics research

TEACHING EXPERIENCE

Principles of Functional Programming [[Website](#)]

May 2021 – May 2026

Course Instructor

May 2025 – Aug 2025

- Developed and presented lectures on topics in functional programming to a class of 40 students
- Authored quizzes and exams, recitation lesson plans, and course resources

Head Teaching Assistant

Jan 2023 – Dec 2023

- Managed and provided guidance for a course of 400 students and a staff of 40 TAs
- Led recitations, review sessions, and office hours on topics in functional programming
- Developed homework problems, exam problems, and modular frameworks for autograders in **SML**

Teaching Assistant (Infrastructure)

Aug 2025 – May 2026

- Investigated transfer of course material to **OCaml** through assignment development and playtesting
- Developed an autograding infrastructure in **OCaml** using the **QCheck** library

Foundations of Programming Languages [[Website](#)]

Jan 2024 – May 2026

Head Teaching Assistant

Aug 2024 – May 2025

- Led recitations, review sessions, and office hours on topics in programming languages and type theory
- Authored recitation notes, exam practice problems, and homework problems for a course of 40 students

Hype for Types [[Website](#)]

Jan 2023 – May 2025

Course Instructor

- Taught a 20 student credit-bearing course through the CMU Student College (StuCo) on topics in type theory, programming language theory, and constructive logic(s)

PROFESSIONAL EXPERIENCE

CMU CS Academy [[Website](#)]

Oct 2025 – Aug 2026

Software Engineering Intern

- Implemented frontend changes in **React** and **Redux** for the CS Academy online learning platform, including improving search result display and site option accessibility
- Improved backend data collection using the **Django** framework in **Python**
- Developed **Rust**-based library for Python graphics framework

PUBLICATIONS

Conference Publications

TBD Pedestrian Data Collection:

Towards Rich, Portable, and Large-Scale Natural Pedestrian Data

2024

Allan Wang, Daisuke Sato, Yasser Corzo, **Sonya Simkin**, Abhijat Biswas, Aaron Steinfeld

IEEE International Conference on Robotics and Automation (ICRA)

<https://arxiv.org/abs/2309.17187>

Manuscripts

Mechanizing a Proof-Relevant Logical Relation for Timed Message-Passing Protocols

2025

Tesla Zhang, Asher Kornfeld, Rui Li, **Sonya Simkin**, Yue Yao, Stephanie Balzer

<https://arxiv.org/pdf/2511.19521>

A Language-Agnostic Logical Relation for Message-Passing Protocols

2025

Tesla Zhang, **Sonya Simkin**, Rui Li, Yue Yao, Stephanie Balzer

<https://arxiv.org/pdf/2506.10026>

TALKS

Introduction to Call-By-Push-Value

Oct 2024

CMU PLunch, <https://hypefortypes.github.io/F24/lectures/lec08.pdf>

HONORS AND AWARDS

Mark J. Stehlik Introductory and Service Teaching Award

May 2026

School of Computer Science, Carnegie Mellon University

Senior Leadership Recognition

May 2024

Office of Student Activities, Carnegie Mellon University